GAMLSS: Location, Scale, and Shape Models

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Introduction

A standard GAM models the **mean** of a response distribution as a smooth function of covariates, keeping other distribution parameters (variance, shape) constant. **GAMLSS** (Generalized Additive Models for Location, Scale, and Shape) extends this by allowing *every* parameter of the response distribution to depend on covariates through smooth functions.

For a response y_i with distribution $\mathcal{D}(\theta_1, \theta_2, \ldots, \theta_K)$, GAMLSS models each parameter via its own additive predictor:

$$g_k(\theta_{ki}) = \beta_{0k} + f_{k1}(x_{1i}) + \cdots + f_{kJ_k}(x_{J_k i}), \quad k = 1, \ldots, K$$

where g_k is the link function for parameter k. This is useful when:

- **Variance changes** with covariates (heteroscedasticity)
- **Skewness or tail behavior** varies across the covariate space
- You need a **distributional regression** rather than a mean regression

GAM.jl's `gamlss()` function provides this capability using Distributions.jl types and GLM.jl link functions.

Setup

```
using GAM
using CSV
using DataFrames
using Distributions
using GLM: LogLink, LogitLink, IdentityLink
using Statistics: mean, std, var, cor
using StatsAPI: fitted, deviance
using LinearAlgebra: diag
using Random
using Plots
```

Normal Location-Scale Model

The simplest GAMLSS: a Gaussian response where both the mean $\mu(x)$ and standard deviation $\sigma(x)$ vary smoothly with a covariate.

Data

We use data with:

- Location: $\mu(x) = 2 + 1.5\sin(x)$
- Scale: $\sigma(x) = 0.3 + 0.2\cos(x)$

```
dat = CSV.read("data_normal_ls.csv", DataFrame)
println("n = $(nrow(dat)), y range: [$(round(minimum(dat.y); digits=2)), $(round(maximum(dat
```

```
n = 500, y range: [-0.33, 4.21]
```

Fitting the model

With `gamlss()`, pass `Normal()` directly — its parameters (μ, σ) match the Distributions.jl parameterization. Each parameter gets its own formula:

```
m = gamlss(
    [@gam_formula(y ~ s(x, k=15, bs=:cr)),   #  formula
     @gam_formula(y ~ s(x, k=10, bs=:cr))],   # log( ) formula
    dat, Normal()
)
println("Converged: $(m.converged)")
println("Negative log-likelihood: $(round(m.nll; digits=2))")
println("Total EDF: $(round(sum(m.edf); digits=1))")
```

```
Converged: true
Negative log-likelihood: 25.99
Total EDF: 18.6
```

The default links are `IdentityLink()` for μ and `LogLink()` for σ, matching the standard GAMLSS parameterization.

Extracting fitted parameters

The fitted linear predictors are stored in `m.fitted_eta`. Apply the inverse link to recover the distribution parameters:

```
 _fit = m.fitted_eta[1]                    # identity link → directly
 _fit = exp.(m.fitted_eta[2])              # log link → exp to get

println(" : cor with truth = $(round(cor( _fit, dat.mu_true); digits=5))")
println(" : cor with truth = $(round(cor( _fit, dat.sigma_true); digits=5))")
```

```
 : cor with truth = 0.99861
 : cor with truth = 0.98639
```

Comparison with standard GAM

A standard `gam()` assumes constant variance. Let's compare:

```
m_gam = gam(@gam_formula(y ~ s(x, k=15, bs=:cr)), dat)
 _gam = m_gam.fitted_values
println("Standard GAM  : cor = $(round(cor( _gam, dat.mu_true); digits=5))")
println("GAMLSS  :       cor = $(round(cor( _fit, dat.mu_true); digits=5))")
println("Max | _gam -  _gamlss|: $(round(maximum(abs.( _gam -  _fit)); digits=4))")
```

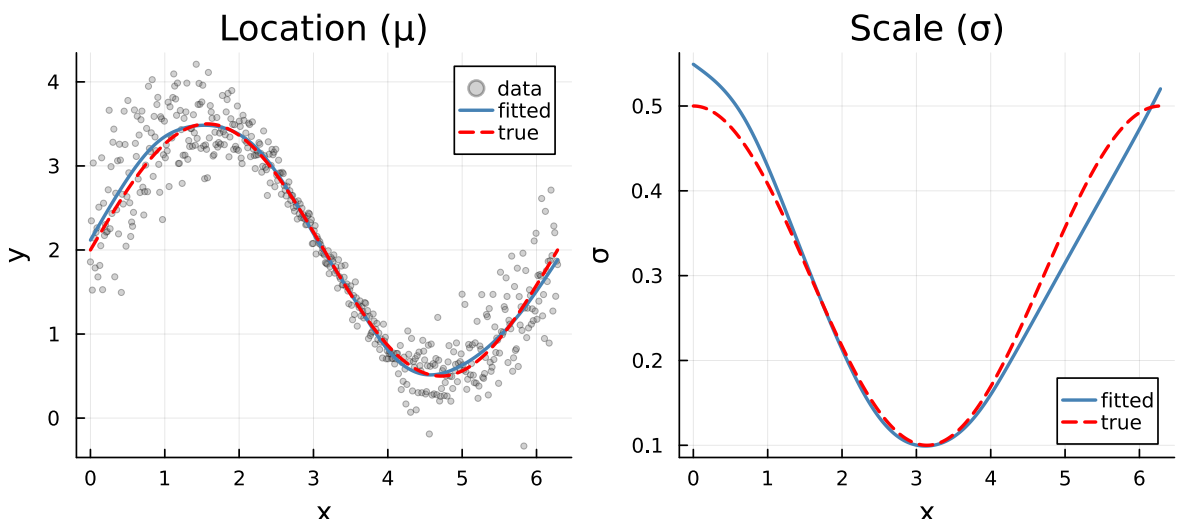
```
Standard GAM  : cor = 0.99873
GAMLSS  :       cor = 0.99861
Max | _gam -  _gamlss|: 0.0357
```

The location estimates are similar, but GAMLSS additionally captures how the spread changes.

Visualisation

```
idx = sortperm(dat.x)

p1 = scatter(dat.x, dat.y; label="data", ms=2, alpha=0.3, color=:grey40,
    xlabel="x", ylabel="y", title="Location ( )")
plot!(p1, dat.x[idx],  _fit[idx]; label="fitted", lw=2, color=:steelblue)
plot!(p1, dat.x[idx], dat.mu_true[idx]; label="true", lw=2, ls=:dash, color=:red)

p2 = plot(dat.x[idx],  _fit[idx]; label="fitted", lw=2, color=:steelblue,
    xlabel="x", ylabel=" ", title="Scale ( )")
plot!(p2, dat.x[idx], dat.sigma_true[idx]; label="true", lw=2, ls=:dash, color=:red)
```

```
plot(p1, p2; layout=(1, 2), size=(700, 300))
```



Constant-variance special case

When σ is modeled as intercept-only (`y ~ 1`), GAMLSS reduces to a standard GAM:

```
m_const = gamlss(
    [@gam_formula(y ~ s(x, k=15, bs=:cr)),
     @gam_formula(y ~ 1)],
    dat, Normal()
)
 _const = exp.(m_const.fitted_eta[2])
println("  range: [$(round(minimum( _const); digits=4)), $(round(maximum( _const); digits=4))]
println("  CV: $(round(std( _const)/mean( _const)*100; digits=2))%")
```

```
  range: [0.3294, 0.3294]
  CV: 0.0%
```

Gamma Location-Scale Model

For positive continuous data, `GammaLocationScale()` reparameterizes the Gamma distribution as:

- μ = mean (link: `LogLink()`)

- σ = coefficient of variation (link: `LogLink()`)

This maps to `Gamma( = 1/ ², = ²)` so that $E[Y] = \mu$ and $\text{Var}[Y] = \mu^2\sigma^2$.

Data

```
dat_g = CSV.read("data_gamma_ls.csv", DataFrame)
println("n = $(nrow(dat_g)), y range: [$(round(minimum(dat_g.y); digits=3)), $(round(maximum
println("All positive: $(all(dat_g.y .> 0))")
```

```
n = 500, y range: [0.563, 7.16]
All positive: true
```

Fitting

```
m_g = gamlss(
    [@gam_formula(y ~ s(x, k=15, bs=:cr)),   # log( )
     @gam_formula(y ~ s(x, k=10, bs=:cr))],   # log( )
    dat_g, GammaLocationScale()
)
println("Converged: $(m_g.converged)")
```

```
Converged: true
```

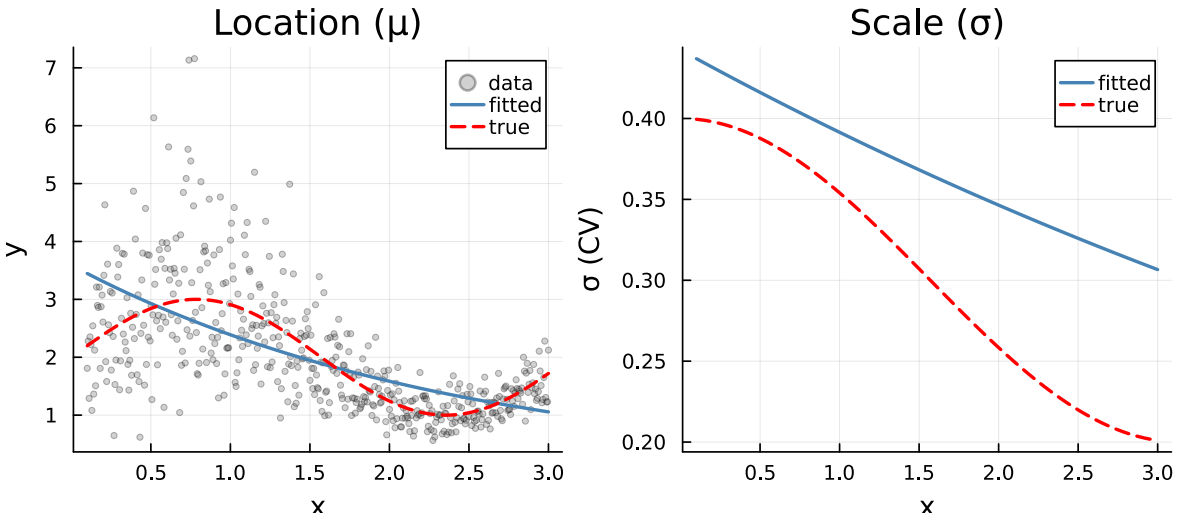
Results

```
 _fit = exp.(m_g.fitted_eta[1])   # log link for
 _fit = exp.(m_g.fitted_eta[2])   # log link for   (CV)

println(" : cor with truth = $(round(cor( _fit, dat_g.mu_true); digits=5))")
println(" : cor with truth = $(round(cor( _fit, dat_g.sigma_true); digits=5))")
println("Mean fitted CV: $(round(mean( _fit); digits=3))")
```

```
 : cor with truth = 0.83194
 : cor with truth = 0.99327
Mean fitted CV: 0.368
```

Visualisation

```
idx_g = sortperm(dat_g.x)

p1 = scatter(dat_g.x, dat_g.y; label="data", ms=2, alpha=0.3, color=:grey40,
    xlabel="x", ylabel="y", title="Location ( )")
plot!(p1, dat_g.x[idx_g],  _fit[idx_g]; label="fitted", lw=2, color=:steelblue)
plot!(p1, dat_g.x[idx_g], dat_g.mu_true[idx_g]; label="true", lw=2, ls=:dash, color=:red)

p2 = plot(dat_g.x[idx_g],  _fit[idx_g]; label="fitted", lw=2, color=:steelblue,
    xlabel="x", ylabel="  (CV)", title="Scale ( )")
plot!(p2, dat_g.x[idx_g], dat_g.sigma_true[idx_g]; label="true", lw=2, ls=:dash, color=:red)

plot(p1, p2; layout=(1, 2), size=(700, 300))
```



Why not just `Gamma()` in `gam()`?

A standard `gam(...; family=Gamma())` only models the mean with a single scale parameter. The GAMLSS version lets the CV *vary* with x, capturing heterogeneity in the dispersion.

Beta Regression

For responses bounded in $(0, 1)$ — proportions, rates, or compositional data — `BetaRegression()` reparameterizes the Beta distribution as:

- μ = mean (link: `LogitLink()`)
- ϕ = precision (link: `LogLink()`)

This maps to `Beta( = \mu\phi, = (1-\mu)\phi)` so that $E[Y] = \mu$ and $\text{Var}[Y] = \mu(1-\mu)/(1+\phi)$.

Data

```
dat_b = CSV.read("data_beta_reg.csv", DataFrame)
println("n = $(nrow(dat_b)), y range: [$(round(minimum(dat_b.y); digits=4)), $(round(maximum
```

```
n = 400, y range: [0.0071, 0.7526]
```

Fitting

Here ϕ is constant (intercept-only), so we model only the mean as smooth:

```
m_b = gamlss(
    [@gam_formula(y ~ s(x, k=15, bs=:cr)),   # logit( )
     @gam_formula(y ~ 1)],                     # log( )
    dat_b, BetaRegression()
)
println("Converged: $(m_b.converged)")
```

```
Converged: true
```

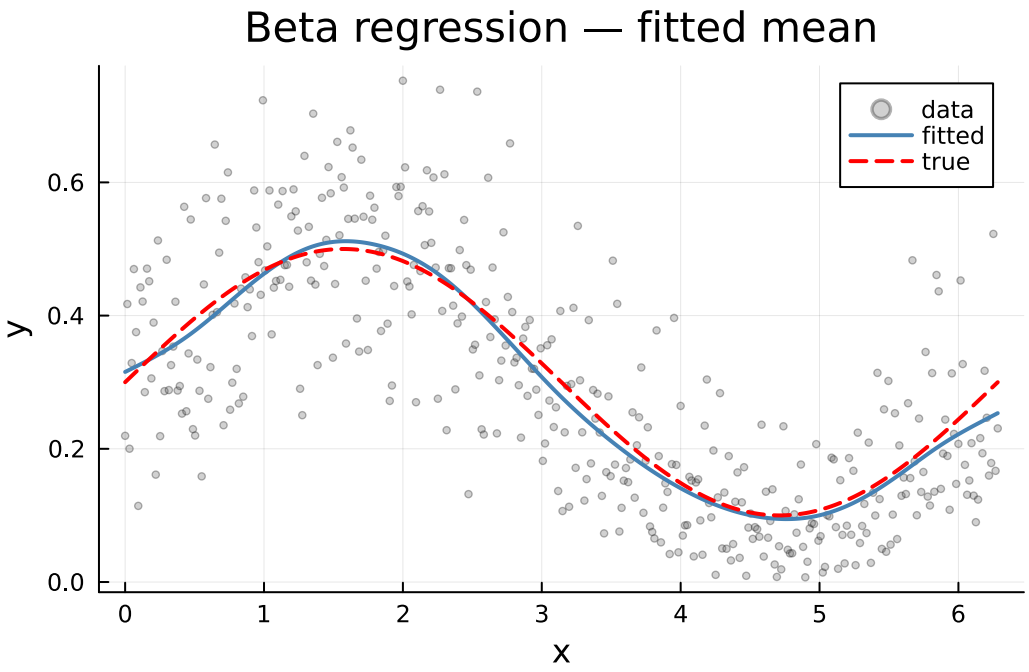
Results

```
logit_inv(x) = 1 / (1 + exp(-x))
 _fit = logit_inv.(m_b.fitted_eta[1])
 _fit = exp.(m_b.fitted_eta[2])

println(" : cor with truth = $(round(cor( _fit, dat_b.mu_true); digits=5))")
println("Fitted precision    $(round(mean( _fit); digits=1)) (true: 15.0)")
```

```
 : cor with truth = 0.99727
Fitted precision    16.5 (true: 15.0)
```

Visualisation

```
idx_b = sortperm(dat_b.x)

scatter(dat_b.x, dat_b.y; label="data", ms=2, alpha=0.3, color=:grey40,
    xlabel="x", ylabel="y", title="Beta regression - fitted mean")
plot!(dat_b.x[idx_b],  _fit[idx_b]; label="fitted", lw=2, color=:steelblue)
plot!(dat_b.x[idx_b], dat_b.mu_true[idx_b]; label="true", lw=2, ls=:dash, color=:red)
```



Custom Links

Links are specified separately from the family, just like in GLM.jl. For `Normal()`, pass them via the `links` keyword:

```
# Use log link for both   and   (e.g., when   must be positive)
Random.seed!(123)
x_pos = collect(range(0.1, 3.0; length=200))
 _pos = exp.(0.5 .+ 0.3 .* sin.(2 .* x_pos))
y_pos =  _pos .+ 0.2 .* randn(200)
y_pos = max.(y_pos, 0.01)
dat_pos = DataFrame(x=x_pos, y=y_pos)
```

```
m_custom = gamlss(
    [@gam_formula(y ~ s(x, k=10, bs=:cr)),
     @gam_formula(y ~ 1)],
    dat_pos, Normal();
    links=[LogLink(), LogLink()]  # both parameters on log scale
)
println("Converged: $(m_custom.converged)")
println("Fitted   range: [$(round(minimum(exp.(m_custom.fitted_eta[1])); digits=2)), $(round(
```

```
Converged: true
Fitted   range: [1.26, 2.17]
```

For reparameterized families, pass links to the constructor:

```
m_g2 = gamlss(
    [@gam_formula(y ~ s(x, k=10, bs=:cr)),
     @gam_formula(y ~ 1)],
    dat_g, GammaLocationScale(links=[LogLink(), LogLink()])
)
println("Converged: $(m_g2.converged)")
```

```
Converged: true
```

Available Families

GAM.jl's `gamlss()` supports several distribution families:

Family	Parameters	Default links	Type
`Normal()`	(mean), (SD)	Identity, Log	Direct Distributions.jl
`GammaLocationScale()`	(mean), (CV)	Log, Log	Reparameterized
`BetaRegression()`	(mean), (precision)	Logit, Log	Reparameterized
`NegativeBinomialLocationScale()`	(mean), (overdispersion)	Log, Log	Reparameterized
`InverseGaussianLocationScale()`	(mean), (CV)	Log, Log	Reparameterized

Family	Parameters	Default links	Type
`GEVFamily()`	, ,	Identity, Log, Identity	Extreme value
`GPDFamily()`	,	Log, Identity	Extreme value

Legacy aliases (`GaussianLS()`, `GammaLS()`, `BetaLS()`, `NegBinLS()`) are also available for backward compatibility.

Effective Degrees of Freedom

Each smooth in each parameter equation has its own EDF, estimated via REML:

```
println("Normal location-scale model:")
K = GAM.nparams(m.family)
offsets = m.param_offsets
for k in 1:K
    s = offsets[k] + 1
    e = offsets[k + 1]
    edf_k = sum(m.edf[s:e])
    pname = GAM.param_names(m.family)[k]
    println("  $(pname): total EDF = $(round(edf_k; digits=2)) ($(e-s+1) coefficients)")
end
```

```
Normal location-scale model:
  mu: total EDF = 11.51 (15 coefficients)
  sigma: total EDF = 7.07 (10 coefficients)
```

R Comparison

The R `gamlss` package (with `gamlss.add` for smooth terms) fits similar models. Here we verify GAM.jl matches R:

```
using RCall

# Re-fit the Normal location-scale model for comparison
m_norm = gamlss(
    [@gam_formula(y ~ s(x, k=15, bs=:cr)),
     @gam_formula(y ~ s(x, k=10, bs=:cr))],
```

```
    dat, Normal())
_fit_norm = m_norm.fitted_eta[1]
_fit_norm = exp.(m_norm.fitted_eta[2])

R"""
suppressPackageStartupMessages({
    library(mgcv)
    library(gamlss)
    library(gamlss.add)
})
"""

# Pass data to R and fit Normal location-scale
@rput dat
R"""
m_r <- gamlss(y ~ ga(~s(x, k=15, bs="cr")),
              sigma.formula = ~ ga(~s(x, k=10, bs="cr")),
              family = NO(), data = dat, trace = FALSE,
              control = gamlss.control(n.cyc = 100, trace = FALSE))
mu_r <- fitted(m_r, "mu")
sigma_r <- fitted(m_r, "sigma")
"""
@rget mu_r sigma_r

n_julia = length( _fit_norm)
n_r = length(mu_r)
if n_julia == n_r
    println(" : Julia-R correlation = $(round(cor( _fit_norm, mu_r); digits=6))")
    println(" : Julia-R correlation = $(round(cor( _fit_norm, sigma_r); digits=6))")
    println(" : max |diff| = $(round(maximum(abs.( _fit_norm - mu_r)); digits=4))")
    println(" : max |diff| = $(round(maximum(abs.( _fit_norm - sigma_r)); digits=4))")
else
    println("Length mismatch: Julia=$n_julia, R=$n_r - comparing first $n_r")
     _j =  _fit_norm[1:n_r]
     _j =  _fit_norm[1:n_r]
    println(" : Julia-R correlation = $(round(cor( _j, mu_r); digits=6))")
    println(" : Julia-R correlation = $(round(cor( _j, sigma_r); digits=6))")
end
```

Summary

- `gamlss()` fits distributional regression models where each parameter gets its own smooth predictor
- Pass **Distributions.jl types** directly (`Normal()`) when parameters match, or use **reparameterized types** (`GammaLocationScale()`, `BetaRegression()`) when they don't
- **Links** are always separate — via `links=` keyword for direct types, or in the family constructor for reparameterized types
- An intercept-only scale formula (`y ~ 1`) recovers the standard GAM as a special case
- Smoothing parameters are estimated jointly via **REML**
- Results can be compared against R's `gamlss` package with `gamlss.add` smooth terms